$\mathrm{rank}(A) = r$

$P^{-1}AP = D$

$A\mathbf{x} = \mathbf{b}$

$A\mathbf{v} = \lambda\mathbf{v}$

$P^{-1}AP = D$

$A\mathbf{x} = \mathbf{b}$

$\det(A - \lambda I) = 0$

$\mathrm{tr}(A) = \sum \lambda_i$

$\det(A - \lambda I) = 0$

det

λ

$\mathrm{tr}(A) = \sum \lambda_i$

$A\mathbf{v} = \lambda\mathbf{v}$

$A^{-1} = \frac{1}{\det A}\,\mathrm{adj}(A)$

$\mathbf{v}$

A

$A^{-1} = \frac{1}{\det A}\,\mathrm{adj}(A)$

$\mathrm{rank}(A) = r$